```java
//SIT 55 : ASSIGNMENT 11 : STRATEGY DESIGN PATTERN
/*11.Implement and apply Strategy Design pattern for simple Shopping Cart where three payment
strategies are used such as Credit Card, PayPal, Bit Coin. Create an interface for strategy pattern
and give concrete implementation for payment*/

import java.util.Scanner;
interface PaymentProcessor
{
 void pay(int amount);
}
class CreditCard implements PaymentProcessor
{
 Scanner sc =new Scanner (System.in);
 String name,ExpDate;
 double CardNo;
 CreditCard()
 {
 super();
 System.out.println("---------------------------------------------------------");
 System.out.print("\tCard holder Name :: ");
 this.name =sc.next();
 System.out.print("\tCard Number :: ");
 this.CardNo =sc.nextDouble();
 System.out.print("\tCard Expire Date :: ");
 this.ExpDate =sc.next();
 System.out.println("---------------------------------------------------------");
 }
 //@Override
 public void pay(int amount)
 {
 System.out.println("---------------------------------------------------------");
 System.out.println("Paying through CreditCard payment: Charging $" + amount);
 System.out.println("---------------------------------------------------------");
 }

}

class PayPal implements PaymentProcessor
{
 PayPal()
 {
  super();
  System.out.println("\nChecking Internet Connection........");
 }
 //@Override
 public void pay(int amount)
 {
  System.out.println("---------------------------------------------------------");
  System.out.println("Paying through PayPal payment: Charging $" + amount);
  System.out.println("---------------------------------------------------------");
 }
}

class BitCoin implements PaymentProcessor
{
```

```
 Scanner sc =new Scanner (System.in);
 String add;
 BitCoin()
 {
 super();
 System.out.print("\nEnter Transaction 'Input Address' :: ");
 add= sc.next();
 }
 //@Override
 public void pay(int amount)
 {
 System.out.println("-----------------------------------------------------------");
 System.out.println("Paying through BitCoin payment: Charging $" + amount);
 System.out.println("-----------------------------------------------------------");
 }
}

class Order
{
 private final PaymentProcessor paymentProcessor;
 private final int amount;
 public Order(int amount, PaymentProcessor paymentProcessor)
 {
 this.amount = amount;
 this.paymentProcessor = paymentProcessor;
 }
 public void process()
 {
 paymentProcessor.pay(amount);
 }
}

public class Main
{
 public static void main(String[] args)
 {
  int c,amt=0;
  Order order;
  Scanner sc = new Scanner(System.in);
  while(true)
  {
   System.out.println();
   System.out.println("**** SHOPPING CART ****");
   System.out.print("1.Credit Card \n2.PayPal \n3.BitCoin \n4.Exit");
   System.out.print("\n\nEnter the Choice: ");
   c=sc.nextInt();
   System.out.println("-----------------------------------------------------------");
   if(c==1||c==2||c==3)
   {
    System.out.print("\nEnter amount to be Tranfer: ");
    amt = sc.nextInt();
    System.out.println("-----------------------------------------------------------");
   }
   switch(c)
   {
```

```
    case 1:
     order = new Order(amt, new CreditCard());
     order.process();
    break;
    case 2:
     order = new Order(amt, new PayPal());
     order.process();
    break;
    case 3:
     order = new Order(amt, new BitCoin());
     order.process();
    break;
    case 4:
    System.out.println("\nThank you For Shopping !!!! ");
    System.out.println("-------------------------------------------------------");
    return;
    default:
    System.out.println("Invalid Payment Mode !!!");
    System.out.println("-------------------------------------------------------");
   }
  }
 }
}


/*OUTPUT:
**** SHOPPING CART ****
1.Credit Card
2.PayPal
3.BitCoin
4.Exit

Enter the Choice: 1
-------------------------------------------------------

Enter amount to be Tranfer: 3300
-------------------------------------------------------
-------------------------------------------------------
	Card holder Name :: ABC
	Card Number :: 551155
	Card Expire Date :: 09/2024
-------------------------------------------------------
-------------------------------------------------------
Paying through CreditCard payment: Charging $3300
-------------------------------------------------------

**** SHOPPING CART ****
1.Credit Card
2.PayPal
3.BitCoin
4.Exit

Enter the Choice: 2
-------------------------------------------------------
```

```
Enter amount to be Tranfer: 2200
--------------------------------------------------------

Checking Internet Connection........
--------------------------------------------------------
Paying through PayPal payment: Charging $2200
--------------------------------------------------------

**** SHOPPING CART ****
1.Credit Card
2.PayPal
3.BitCoin
4.Exit

Enter the Choice: 3
--------------------------------------------------------

Enter amount to be Tranfer: 10000
--------------------------------------------------------

Enter Transaction ’Input Address’ :: AABBCC
--------------------------------------------------------
Paying through BitCoin payment: Charging $10000
--------------------------------------------------------

**** SHOPPING CART ****
1.Credit Card
2.PayPal
3.BitCoin
4.Exit

Enter the Choice: 4
--------------------------------------------------------

Thank you For Shopping !!!!
--------------------------------------------------------*/
```